

2-Digit Digital Stopwatch

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Abstract

To design and implement a 2-Digit Digital Stopwatch on the real digital boolean board. The output will be displayed both in binary and in decimal using leds and the seven segment display on the fpga board.

1 Stop Watch

1.1 Introduction

This assignment requires:

- Counting from **00 to 59** seconds with automatic rollover to 00.
- Displaying the count in **binary** via on-board LEDs.
- Displaying the count in **decimal** using the seven segment display on the fpga board.
- A **Start/Stop** push-button that toggles between running and paused states.
- A **Reset** push-button that immediately resets the counter to 00.

1.2 Verilog Code

```
1 module stopwatch (
2     input wire clk,
3     input wire start_stop,
4     input wire reset_btn,
5     output wire [5:0] led,
6     output reg [7:0] anode,
7     output reg [7:0] seg
8 );
9
10 // -- 1 Hz tick -----
11 reg [26:0] cnt_1hz;
12 wire tick_1hz = (cnt_1hz == 27'd99_999_999);
13
14 always @(posedge clk)
15     cnt_1hz <= tick_1hz ? 27'd0 : cnt_1hz + 1;
16
17 // -- 1 kHz display mux tick -----
18 reg [16:0] cnt_mux;
19 wire tick_mux = (cnt_mux == 17'd99_999);
20
21 always @(posedge clk)
22     cnt_mux <= tick_mux ? 17'd0 : cnt_mux + 1;
23
24 // -- BCD counter -----
25 reg [3:0] sec_units;
26 reg [3:0] sec_tens;
27
28 always @(posedge clk) begin
29     if (reset_btn) begin
30         sec_units <= 4'd0;
31         sec_tens <= 4'd0;
32     end else if (tick_1hz && start_stop) begin
33         if (sec_units == 4'd9) begin
34             sec_units <= 4'd0;
35             sec_tens <= (sec_tens == 4'd5) ? 4'd0 : sec_tens
36 + 1;
37         end else begin
38             sec_units <= sec_units + 1;
39         end
40     end
41 end
42
43 // -- Binary LEDs -----
44 assign led = sec_tens * 6'd10 + sec_units;
```

```

44
45 // -- 7-segment decoder (active LOW) -----
46 // Bit order: {dp, g, f, e, d, c, b, a}
47 function [7:0] seg7;
48     input [3:0] d;
49     case (d)
50         4'd0: seg7 = 8'b11000000;
51         4'd1: seg7 = 8'b11111001;
52         4'd2: seg7 = 8'b10100100;
53         4'd3: seg7 = 8'b10110000;
54         4'd4: seg7 = 8'b10011001;
55         4'd5: seg7 = 8'b10010010;
56         4'd6: seg7 = 8'b10000010;
57         4'd7: seg7 = 8'b11111000;
58         4'd8: seg7 = 8'b10000000;
59         4'd9: seg7 = 8'b10010000;
60         default: seg7 = 8'b11111111;
61     endcase
62 endfunction
63
64 // -- Multiplexed display -----
65 reg disp_sel;
66
67 always @(posedge clk) begin
68     if (tick_mux) begin
69         disp_sel <= ~disp_sel;
70         if (disp_sel == 1'b0) begin
71             anode <= 8'b11101111; // units digit ON
72             seg <= seg7(sec_units);
73         end else begin
74             anode <= 8'b11011111; // tens digit ON
75             seg <= seg7(sec_tens);
76         end
77     end
78 end
79
80 endmodule

```

1.3 Hardware Mapping

Signal Name	Component Type	Board Label	FPGA Pin
clk	System Clock	Internal	F14
start_stop	Slide Switch	SW0	V2
reset_btn	Push Button	BTN0	J2
led[5]	LED (MSB)	LD5	E2
led[4]	LED	LD4	E1
led[3]	LED	LD3	F2
led[2]	LED	LD2	F1
led[1]	LED	LD1	G2
led[0]	LED (LSB)	LD0	G1
seg[7]	7-Seg Cathode	DP	B7
seg[6]	7-Seg Cathode	CG	A7
seg[5]	7-Seg Cathode	CF	C5
seg[4]	7-Seg Cathode	CE	D7
seg[3]	7-Seg Cathode	CD	A8
seg[2]	7-Seg Cathode	CC	C7
seg[1]	7-Seg Cathode	CB	C4
seg[0]	7-Seg Cathode	CA	D5
anode[1]	7-Seg Anode	AN1 (Tens Digit)	J4
anode[0]	7-Seg Anode	AN0 (Units Digit)	H3

1.4 Functional Blocks in Code

1.4.1 Clock Dividers

The design utilizes a single 100 MHz master clock to generate two critical, slower timebases via counters:

- **1 Hz Tick:** A 27-bit counter (`cnt_1hz`) counts to 99,999,999 to generate a precise 1-second pulse. This drives the actual timekeeping of the stopwatch.
- **1 kHz Tick:** A 17-bit counter (`cnt_mux`) counts to 99,999 to generate a 1 millisecond pulse. This is used to refresh the multiplexed 7-segment display fast enough that the human eye perceives both digits as being on simultaneously.

1.4.2 BCD Counter

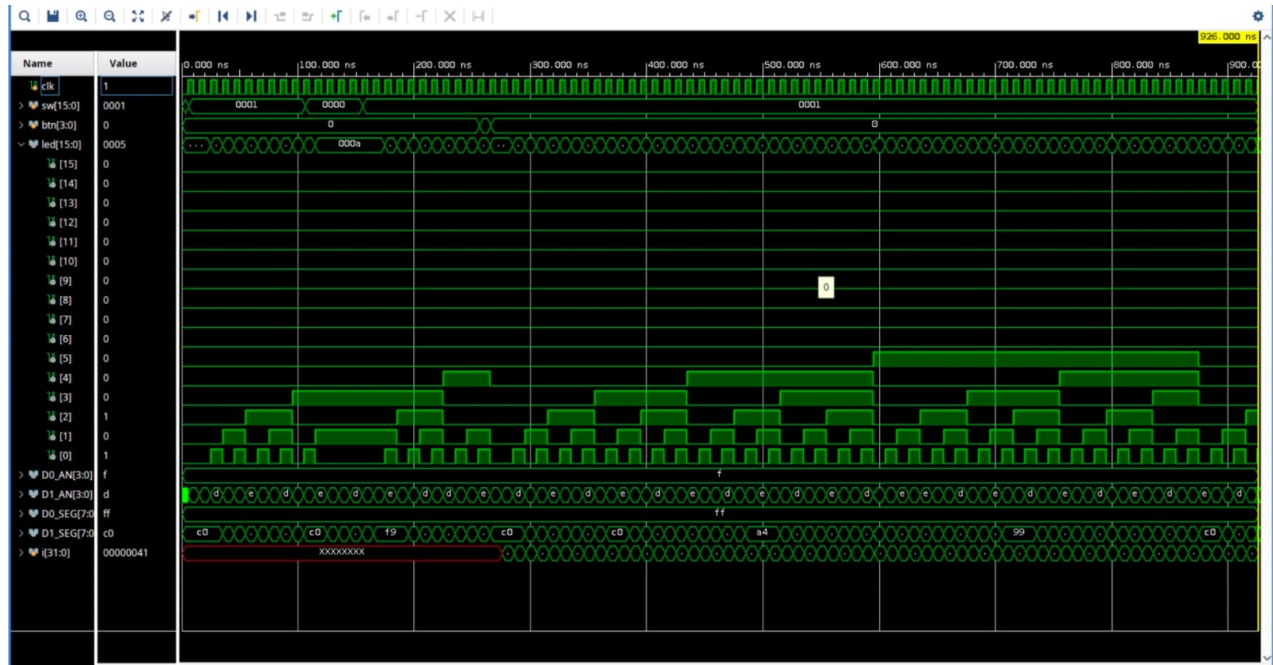
Instead of a single binary counter, the time is tracked using Binary-Coded Decimal (BCD) with separate registers for `sec_units` (0-9) and `sec_tens` (0-5).

- When the 1 Hz tick occurs (and the stopwatch is running), `sec_units` increments.
- When `sec_units` rolls over from 9 to 0, `sec_tens` increments.
- When `sec_tens` reaches 5 and `sec_units` is at 9 (59 seconds), both reset to 0 on the next tick.

1.5 Output

- **Binary LEDs:** The BCD values are converted back to a flat binary value via the equation $(\text{sec_tens} * 10) + \text{sec_units}$ to drive the 6-bit LED output.
- **7-Segment Display:** A lookup function (`seg7`) translates the 4-bit BCD digit into an active-low 8-bit cathode pattern. The `cnt_mux` tick toggles a selection bit (`disp_sel`), rapidly switching power (via the anode register) back and forth between the tens digit and the units digit while feeding the corresponding cathode pattern to `seg`.
- We have demonstrated the entire process in two videos.
- 1st video consists of Running synthesis, implementation, Generating bit stream and flashing it onto FPGA board via hardware manager.
- 2nd video explains the different states and cases of output

1.6 Output Waveform



output waveform

1.7 Output Links

- Please click on the following link which includes the part of flashing code to the fpga board
Programming the FPGA board
- The video below has the demonstration of the output on the FPGA board
2-Digit Digital Stop Watch

1.8 Conclusion

A fully functional 2-Digit Digital stop watch was successfully designed and implemented on the Real Digital Boolean Digital Board. We were successfully able to get output on both leds and also on the seven segment displays